SUNDEEP CHAKLADAR

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EDUCATION

WashU School of Medicine — St. Louis, MO
MD Student | Distinguished Student Scholar — Aug 2024-Present
Expected Graduation: May 2028

Massachusetts Institute of Technology — Cambridge, MA
B.S. in Biology, Minor in Science, Technology, & Society — Aug 2020-May 2024
GPA: 4.9/5.0 | Honors: Phi Beta Kappa

RESEARCH EXPERIENCE

WashU School of Medicine, *Medical Student Researcher* — St. Louis, MO
Advisors: Dr. David Brogan, Dr. Jennifer Strahle, Dr. Pooya Hosseinzadeh — Aug 2024-Present
Developing AI-based pipelines for automated fracture detection, disease progression analysis, clinical risk stratification, and longitudinal recovery assessment.

Lourido Lab, Whitehead Institute for Biomedical Research, *Undergraduate Researcher* — Cambridge, MA
Advisors: Dr. Sebastian Lourido, Dr. Haley Licon, Dr. Aditi Shukla — Aug 2020-May 2024
Built machine learning-based, microscopy image analysis pipelines to characterize toxoplasma gondii's kinome, motility, and life cycle

Brigham and Women's Hospital, *Data Science and Medical Research Intern* — Boston, MA
Advisor: Dr. Christopher Baugh — May-Sept 2023
Extracted clinical data and identified how to efficiently triage febrile cancer patients presenting to the emergency department

Kellis Lab, MIT Computer Science and Artificial Intelligence, *Undergraduate Researcher* — Cambridge, MA
Advisors: Dr. Manolis Kellis, Dr. Lei Hou — Aug 2022-Aug 2023
Developed machine learning models to predict the effects and progression of disease-induced inflammation across various patient populations

WORK EXPERIENCE

ArmorMed, *Founder, CEO*, www.armormedicine.com — St. Louis, MO
Developing a platform for automated joint mobility assessment and longitudinal patient monitoring using computer vision and pose estimation technology. Secured startup funding through local and national entrepreneurship competitions. — Jan 2025-Present

MIT Emergency Medical Services, *EMT-B, Scheduling Officer, Training Officer* — Boston, MA
Leveraged knowledge of emergency medicine to assess, treat, and transport patients. Created shift schedules and organized training to maintain member certification. — Jan 2022-May 2024

MassHealth, *Clinical Research Intern* — Boston, MA
Investigated and contributed to proposals aimed at decreasing health disparities and expanding healthcare access to Massachusetts' undocumented population — May-Sept 2023

Maine Department of Health and Human Services, *Data Analytics Intern* — Augusta, ME
Applied computational methods to analyze the effects of MaineCare on the accessibility of providers, health insurance, and affordable medical aid — Jan-April 2023

PUBLICATIONS

1. **Chakladar S**, Lee O, Hunter DA, Boukind A, Dy C, Brogan DM. AutoSNAP enables automated nerve segmentation and morphometry quantification. Sci Rep. Published online November 10, 2026. doi:10.1038/s41598-026-78196-9.
2. Pereira DE, **Chakladar S**, Harrison NJ, Rust BD, Obey MR, McAndrew CM, Berkes MB, Wilson JL. Measuring Knee Alignment Following Distal Femur Fracture Surgery: A Novel Deep Learning Approach. J Orthop Trauma. Published online October 7, 2026. doi:10.1097/BOT.0000000000003300
3. **Chakladar S**, Pan S, Limbrick O, Pandey M, Halupnik GL, Zhao A, Mahjoub MR, Quirk JD, Nazeri A, Strahle JM. Automated ventricle assessment via three-dimensional anatomical reconstruction (AVA-TAR): A computational toolkit for autonomous lateral ventricle assessment in preclinical hydrocephalus models. Neuroimage. 2026;338:122124. doi:10.1016/j.neuroimage.2026.122124.
4. Joseph K, Vogl S, Galla JT, Yahanda A, Gupta V, Yakdan S, Ruiz-Cardozo M, Plog B, Bui T, Poulin N, **Chakladar S**, Narro-Garcia F, Ng J, Wang M, Benedict B, Brehm S, Hallak H, Barot K, Pallota N, Gulta M, Neuman B, Greenberg J, Ray W, Molina C. The Impact of Global Imbalance on Pelvic Fixation Failure: A Single Institution Matched Case-Control Study. Global Spine J. Published online April 9, 2026. doi:10.1177/21925682261441195
5. **Chakladar S**, Pereira DE, Shariati J, Hosseinzadeh P. Automated Detection of Pediatric Slipped Capital Femoral Epiphysis: A Deep Learning Approach Using Anatomically Informed Attention Guidance. J Pediatr Orthop. Published online April 7, 2026. doi:10.1097/BPO.0000000000003279
6. **Chakladar S**, Davis CI, Sane ES, Dy CJ, Brogan DM. Accuracy and Reliability of Wearable Devices, Image Analysis Software, and Smartphone Apps for Hand Goniometry: A Systematic Review. Hand (N Y). Published online April 6, 2026. doi:10.1177/15589447261433064
7. **Chakladar S**, Pereira DE, Tang N, Siddabattula R, Hosseinzadeh P. Development and Validation of an Automated Pipeline for the Detection of Monteggia Fracture Dislocations in Pediatric Radiographs. J Pediatr Orthop. Published online December 15, 2025. doi:10.1097/BPO.0000000000003196
8. Joseph K, Bui TT, Yahanda AT, Gupta V, Vogl S, Yakdan S, Galla JT, Ruiz-Cardozo MA, Barot K, **Chakladar S**, Poulin N, Challagundla A, Ng J, Krishnan A, Brehm S, Benedict B, Clohisu JC, Pallotta N, Gulta M, Beuman B, Hills J, Kelly M, Hafez D, Greenberg J, Ray W, Molina CA. Mechanical Failures as Predicted by Achieving Local vs Global T4-L1 Hip Axis Goals: A Single Center Experience. Spine (Phila Pa 1976). Published online July 11, 2025. doi:10.1097/BRS.0000000000005450
9. **Chakladar S**, Dy CJ, Brogan DM. Accuracy of the Automated Range of Motion Observer and Reporter Software for Fully Automated Joint Measurement From Patient Videos. J Am Acad Orthop Surg Glob Res Rev. 2025;9(7):e25.00088. Published 2025 Jul 8. doi:10.5435/JAAOSGlobal-D-25-00088
10. Joseph K, Bui T, Yahanda AT, Yakdan S, Vogl S, Cardozo MR, Galla JT, Leatherman Z, Poulin N, **Chakladar S**, Brehm S, Benedict B, Gupta, M, Pallotta N, Hills J, Kelly M, Greenberg J, Neuman B, Ray W, Molina CA.. Validation and clinical application of the ΔC2 pelvic angle - ΔC2 tilt = Δpelvic tilt equation for predicting pelvic tilt in spinal deformity surgery. Neurosurg Focus. 2025;58(6):E6. doi:10.3171/2025.3.FOCUS2554
11. Licon H, Giuliano C, **Chakladar S**, Shallberg L, Waldman B, Hunter C, Lourido S. A positive feedback loop controls Toxoplasma chronic differentiation. Nat Microbiol 8, 889–904 (2023). https://doi.org/10.1038/s41564-023-01358-2
12. Chan AW, Broncel M, Haseley N, **Chakladar S**, Andree E, Herneisen AL, Shortt E, Treeck M, Lourido S. Analysis of CDPK1 targets identifies a trafficking adaptor complex that regulates microneme exocytosis in Toxoplasma. eLife12:RP85654. https://doi.org/10.7554/eLife.85654.1

PRESENTATIONS

1. Novel Computational Approaches for the Study of Toxoplasma Gondii

MIT Biology Undergraduate Research Symposium | Oral Presentation | 2 Feb 2023

ABSTRACTS

1. **Chakladar S**, Pan S, Kumar S, Limbrick O, Halupnik GL, Zhao A, Strahle JM. A Self-Supervised Foundation Model for Preclinical Hydrocephalus Neuroimaging. Joint Section on Pediatric Neurological Surgery 2026 Annual Meeting, San Francisco, CA, December 3-6 2026. (Digital poster)
2. **Chakladar S**, Pan S, Koleske JP, Loe M, Lucey BP, Musiek E, Strahle JM. Age-Specific Cerebrospinal Fluid Total Protein Reference Limits Across the Lifespan: An Indirect Derivation From a Single-Institution Cohort. Joint Section on Pediatric Neurological Surgery 2026 Annual Meeting, San Francisco, CA, December 3-6 2026. (Digital poster)
3. Pereira DE, **Chakladar S**, Harrison N, Rust B, Oben M, McAndrew C, Berkes M, Wilson J. Automated Quantification of Knee Alignment in Distal Femur Fracture Patients Using Post-Operative Radiographs. Orthopedic Trauma Association, Nashville, TN, September 23-26 2026. (Printed poster)
4. **Chakladar S**, Lee O, Hunt D, Boukind A, Dy C, Brogan D. AutoSNAP: a web-based deep learning framework for comprehensive nerve segmentation and morphometry quantification. American Society for Surgery of the Hand, Boston, MA, September 17-19 2026. (Printed poster)
5. Sierra C, **Chakladar S**, Ghoranpour Arani A, Martin BA, Nazeri A, Strahle J. Redefining Chiari I Malformation Using 4D MRI for Quantification of Global Intracranial CSF Flow. Journal of Clinical and Translational Science. May 8 2026;10(Suppl 1):75–76. (Published conference abstract)
6. **Chakladar S**, Pereira DE, Tang N, Siddabattula R, Hosseinzadeh P. Development and Validation of an Automated Pipeline for the Detection of Monteggia Fracture Dislocations in Pediatric Radiographs. Pediatric Orthopedic Society of North America, Orlando, FL, May 6-8 2026. (Printed poster)
7. **Chakladar S**, Pereira DE, Tang N, Siddabattula R, Hosseinzadeh P. Development and Validation of an Automated Pipeline for the Detection of Monteggia Fracture Dislocations in Pediatric Radiographs. American Academy of Orthopaedic Surgeons, New Orleans, LA, March 2-6 2026. (Oral presentation)
8. Pan S, Limbrick O, Koneru S, **Chakladar S**, Strahle JM. Re-evaluating the approach to preclinical neonatal IVH models: insights from longitudinal data obtained from 157 rats. Joint Section on Pediatric Neurological Surgery 2025 Annual Meeting, Memphis, TN, December 2-6 2025. (Oral presentation)
9. **Chakladar S**, Pereira DE, Tang N, Siddabattula R, Hosseinzadeh P. Development and Validation of an Automated Pipeline for the Detection of Monteggia Fracture Dislocations in Pediatric Radiographs. WashU Medicine EXPLORE Symposium, St. Louis, MO, October 15 2025. (Printed poster)
10. **Chakladar S**, Pan S, Halupnik G, Strale J. A Deep Learning Model For Ventricle Segmentation, 3D Reconstruction, And Ventriculomegaly Detection In Rodents. Congress of Neurological Surgeons 2025 Annual meeting, Los Angeles, CA, October 11-15 2025. (Printed poster)
11. Joseph K, Bui T, Yahanda T, et al. Predictors Of T4-L1 Hip Axis Error After ASD Surgery: Understanding Drivers Of T4-L1 Mismatch And L1PA Error. Congress of Neurological Surgeons 2025 Annual meeting, Los Angeles, CA, October 11-15 2025. (Oral presentation)
12. Joseph K, Bui T, Yahanda T, et al. Anterior Thoracic Malalignment Drives Mechanical Failure After ASD Surgery: The Predictive Value Of T4-L1 Hip Axis Error Directionality. Congress of Neurological Surgeons 2025 Annual meeting, Los Angeles, CA, October 11-15 2025. (Printed poster)
13. Joseph K, Bui T, Yahanda T, et al. Extending The T4-L1 Hip Axis Framework: Predictive Values Of Alignment Error In Short And Long Fusions For Adult Spinal Deformity. Congress of Neurological Surgeons 2025 Annual meeting, Los Angeles, CA, October 11-15 2025. (Printed poster)

14. Joseph K, Bui T, Yahanda T, et al. Pelvic Fixation Failure Is Rare And Is Predicted By Construct Magnitude And Residual Deformity: A Matched Cohort Analysis. Congress of Neurological Surgeons 2025 Annual meeting, Los Angeles, CA, October 11-15 2025. (Oral presentation)
15. Hallak H, Connor M, **Chakladar S**, Nassirinia E, Zipfel G, Osbun J. Treatment of Brain Arteriovenous Malformations in Patients With Hereditary Hemorrhagic Telangiectasia: A Tertiary Referral Center Experience. Congress of Neurological Surgeons 2025 Annual meeting, Los Angeles, CA, October 11-15 2025. (Printed poster)
16. Hallak H, Connor M, **Chakladar S**, Nassirinia E, Zipfel G, Osbun J. Treatment of Brain Arteriovenous Malformations in Patients with Hereditary Hemorrhagic Telangiectasia: A Tertiary Referral Center Experience. Society of NeuroInterventional Surgery 2025 Annual Meeting, Nashville, TN, July 14-18, 2025. (Digital Poster)

AWARDS AND HONORS

Skandalaris Venture Competition, Winner – April 2026
Awarded $2,000 for recognition as one of the most innovative ventures among 100+ WashU startup teams

Olin 2-Minute Pitch Competition, 3rd Place – April 2026
Delivered a compelling 2-minute video pitch, placing 3rd out of 130+ teams, earning $750

Holekamp Seed Fund, Recipient – March 2026
Recognized as a promising student-led startup and awarded $1,000 to aid with costs of business formation

IdeaBounce Pitch Competition, Winner – February 2026, April 2026
Awarded top prize of $600 for entrepreneurial pitch presentation and innovation potential

Best Basic Science Poster on the Data Science Track – Congress of Neurological Surgeons – 2025
Awarded for: A Deep Learning Model for Ventricle Segmentation, 3D Reconstruction, And Ventriculomegaly Detection in Rodents

College Sports Communicators, Academic All American – MIT Varsity Men's Tennis – 2023
Recognized for outstanding athletic and academic achievement

Intercollegiate Tennis Association Scholar Athlete – MIT Varsity Men's Tennis – 2022, 2023, 2024
Varsity scholar athlete with a GPA above 3.5